

RESEARCH PAPER & INTERNSHIP PROJECT REPORT

Title: Machine Learning based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Domain: Financial Analytics & Real Estate Market Intelligence

Collaborating Organizations: Unified Mentor & Parcl Co. Limited

1. Executive Summary

Real estate markets consist of diverse buyer groups including individual homebuyers, institutional buyers, and international investors. Without data-driven segmentation, property recommendations remain generic and marketing campaigns lose efficiency. This project implements an unsupervised machine learning model using K-Means and Hierarchical Clustering algorithms on client transaction history to profile buyers and predict investment motivations. The model is deployed as an interactive analytical dashboard built with Streamlit.

2. Problem Statement

Parcl Co. Limited faced challenges due to a lack of automated buyer classification, leading to:

- Inefficient marketing spending
- Generic property recommendations
- Poor investor targeting

3. Data Science Methodology

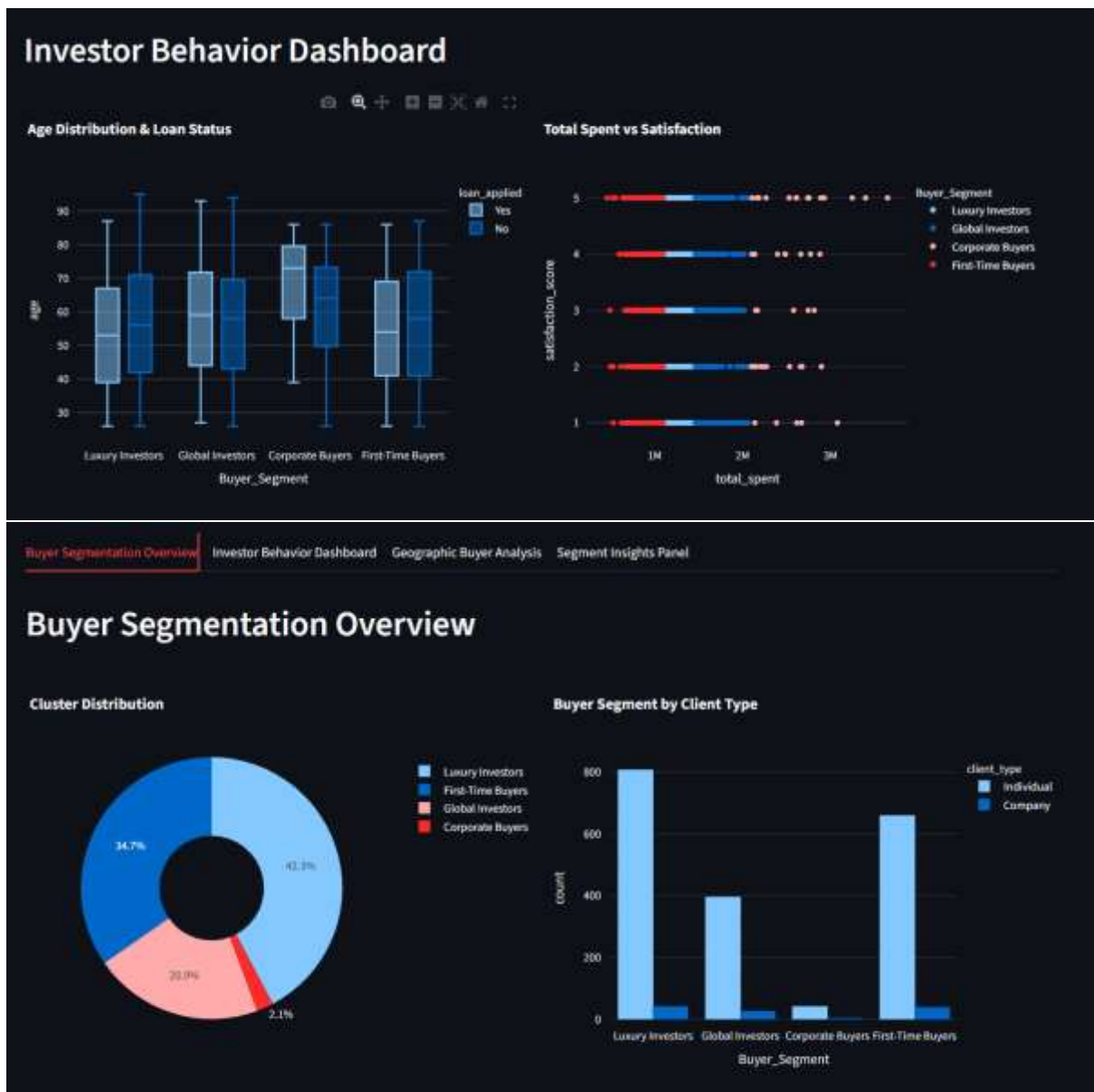
- **Data Integration:** Combined client demographics (clients.csv) with aggregated transaction history (properties.csv) using client_id.
- **Feature Engineering:** Extracted client age from date_of_birth and calculated cumulative spend (total_spent) along with property volume (property_count).
- **Preprocessing:** Applied One-Hot Encoding for categorical features (region, acquisition_purpose, client_type, country) and normalized numeric features using StandardScaler.
- **Clustering Analysis:** Used Elbow Method and Silhouette Analysis to evaluate cluster quality, establishing $K = 4$ as the optimal configuration.

4. Identified Buyer Segments

Cluster	Buyer Segment	Characteristics
C1	Global Investors	High transaction value, cross-border buyers, investment purpose
C2	First-Time Buyers	Younger profile, loan-dependent, personal residence buyers
C3	Corporate Buyers	Institutional entities acquiring multiple units
C4	Luxury Investors	High satisfaction score, premium asset portfolios

5. Streamlit Interactive Dashboard





The deployed web application provides:

1. **Buyer Segmentation Overview:** Cluster proportions and client type breakdowns.
2. **Investor Behavior Dashboard:** Age distribution, loan behavior, and satisfaction mapping.
3. **Geographic Buyer Analysis:** Regional breakdown across client demographics.
4. **Segment Insights Panel:** Descriptive statistics for strategic decision-making.

6. Conclusion

By integrating machine learning into real estate analytics, Parcl can target buyer segments with personalized offerings, lowering customer acquisition costs and improving conversion efficiency.